From Words to Worlds

Milestone 1 and a Comparison of Text Encoders without Training

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1. Summary

This report covers one notebook in two parts. **Part 1** is the complete Milestone 1 pipeline: MS-COCO 2017 validation (5,000 photos, 25,004 captions) cleaned and preprocessed, frozen splits, object labels and stress prompts for later evaluation, the text → embedding pipeline that conditions Stable Diffusion 1.5 (verified three ways), and the first sample images. **Part 2** uses the same 2,000 test captions to compare four Transformer text encoders (CLIP ViT-L/14, BERT-base, T5-base, GPT-2) and a word-count baseline **without training anything**, to give evidence for the project's "embedding choice" question before any connector is trained. The notebook ran end to end on a Tesla T4 with no errors; every Part 1 figure, including the generated images, came out pixel-identical to an earlier independent run, which confirms the pipeline is fully reproducible.

Result	Value	What it shows
Images, captions after cleaning	5,000 · 25,004	The whole validation split; wording never changed
Frozen splits	2,000 test · 500 dev · 2,000 second real set	Fair tuning and a real-photo baseline for FID
Test captions with a scorable object	1,692 of 2,000	Enough targets to measure object recall later
Image from cached embeddings vs raw text	mean pixel difference 0.000	Our pipeline is exactly what drives the model
Caption → photo retrieval (2,000 photos)	R@1 46.8%, R@10 84.0%	Embeddings carry meaning: about 935× chance
Sample runs	5 captions × 3 seeds, 5.1 s per image	Milestone 1 sample deliverable
CKA with Stable Diffusion's CLIP	T5 0.60 · BERT 0.53 · TF-IDF 0.51 · GPT-2 0.13	Text-only encoders organise captions differently from CLIP
Paraphrase retrieval R@1	CLIP (EOS) 34.0% · CLIP 21.4% · TF-IDF 20.0% · T5 15.5% · BERT 13.0% · GPT-2 3.4%	Only CLIP beats plain word counting
Scene clustering (10-NN)	CLIP (EOS) 74.5% · CLIP 70.0% · T5 68.5% · BERT 65.8% · TF-IDF 64.4% · GPT-2 56.0%	All above the 29.9% majority class; CLIP clusters best
Similarity of role-swapped sentences	CLIP 0.91 · T5 0.98 · BERT 0.99 · GPT-2 1.00	Text-only sentence vectors barely change when who-does-what is swapped

Part 1 · Milestone 1

2. Milestone 1 requirements

Requirement	Where	Result	Status
Choose a dataset	Section 2	MS-COCO 2017 val: 5,000 photos, 5 captions each, boxes for 80 object types	Done
Clean and preprocess 1k–5k image–caption pairs	Sections 3–5	5,000 images and 25,004 captions; 512 × 512; every change logged	Done
Set up the environment (PyTorch + Diffusers + CLIP)	Install, Section 1	PyTorch 2.11, Diffusers 0.42.1, Transformers 5.22.1, Tesla T4	Done

Requirement	Where	Result	Status
Baseline notebook for the text → embedding pipeline	Section 10	Cache, inspection, retrieval, same-encoder and parity checks	Done
Sample 5 prompt → image test runs	Section 11	5 captions × 3 seeds beside the real photos, from cached embeddings	Done
1-page proposal (dataset, model, roles)	—	Group document, submitted separately	Group
GitHub repository initialised	—	Notebook and results are ready to push	Group

3. Design choices and why

Every choice either keeps later measurements fair or makes the work reproducible.

Choice	Why
MS-COCO 2017 validation, all 5,000 photos	Each caption has a real photo and labelled object boxes, so we can check whether a generated image shows what the caption names. It is the standard text-to-image benchmark, so results are comparable with published work. Using all photos avoids picking easy ones.
Light caption cleaning: spaces only, wording kept	Changing even a full stop changes the tokens CLIP reads and therefore the image. Real prompts are imperfect, and published results use raw captions.
Center crop to a square, then resize to 512 × 512	Stable Diffusion 1.5 paints at 512 × 512. Stretching wide photos would distort every object and bias image metrics.
PNG, not JPEG	JPEG adds compression marks that generated images do not have; metrics would pick up that difference instead of real quality.
Photos under 256 px kept out of the real reference set	Enlarging them more than twice blurs them, which would unfairly penalise every generated image compared with them.
Rotation tags and data problems counted, never fixed	Object boxes refer to the photo as stored; rotating or removing data would change the test set. Counting still documents data quality.
Frozen 2,000-pair test split, fixed seed	Every later setting is scored on the same captions, so differences come from the setting, and the split is reproducible.
Separate 500-pair development split	Settings are chosen on dev and reported on test. Choosing on the test split makes the winner look better than it is.
Second, disjoint set of 2,000 real photos	FID depends strongly on sample size, so it is only meaningful beside the FID that real photos get against other real photos at the same count.
Object labels with plurals, synonyms, people words, two-word names	To score "did it draw the dog?", the code must know a caption saying "puppies" names a dog, and that "hot dog" does not.
Unclear words excluded; crop check	"An orange cat" contains no fruit, and an object cut out of the real photo by the crop cannot fairly count against the model.
200 stress prompts with expected answers	Natural captions rarely test counting, colour binding, position or word order; stored answers make them checkable by script.
Stable Diffusion's own tokenizer and text encoder, one caption at a time	These are exactly what the model reads. Half-precision GPU arithmetic differs slightly in large batches, and 30 denoising steps amplify that into different images.
Samples from cached embeddings, 3 seeds, beside the real photo	Proves the pipeline end to end, shows what the caption describes, and shows variation across seeds. Defaults (PNDM, 30 steps, guidance 7.5) are a neutral baseline; tuning is Milestone 2.

4. Dataset and caption cleaning

What was done. Downloaded COCO 2017 validation (5,000 photos, 25,014 captions, 36,781 object boxes in 80 categories). Each caption had newlines and repeated spaces collapsed and spaces before punctuation removed; exact duplicate captions of the same image (ignoring case) were dropped.

Outputs. 3,832 captions had spacing repaired and 10 duplicates were dropped, leaving 25,004 captions (4 to 7 per photo). The median caption has 10 words and the longest 49. 48 photos have no labelled object.

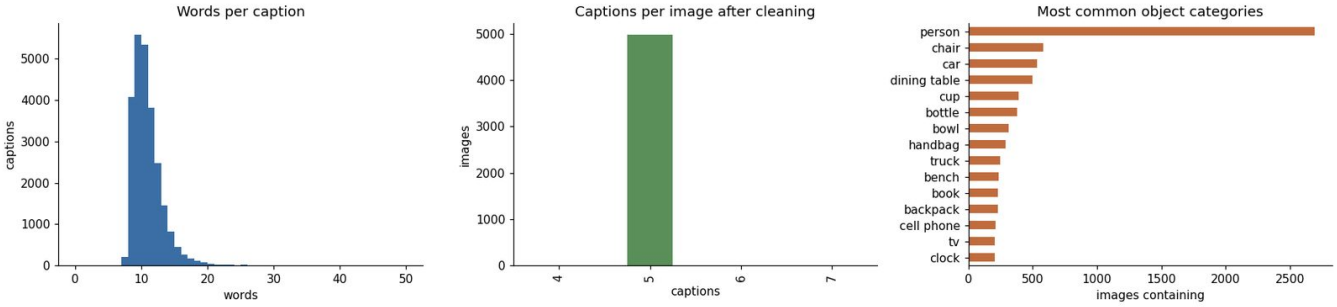


Figure 1. Words per caption, captions per image after cleaning, and the most common object categories. Person dominates, which is why later object-recall results report person separately.

5. Images: 512 × 512

What was done. Every photo was converted to RGB, center-cropped to a square, resized to 512 × 512 with Lanczos filtering and saved as PNG. The same pass recorded size, colour mode, rotation tag and a 256-bit image hash.

Outputs. 10 grayscale photos were converted to RGB. The crop keeps 71.6% of a frame on average (24.8% at worst, for very wide photos). 4,958 photos are eligible for the real reference set; 42 small photos are kept out of it.

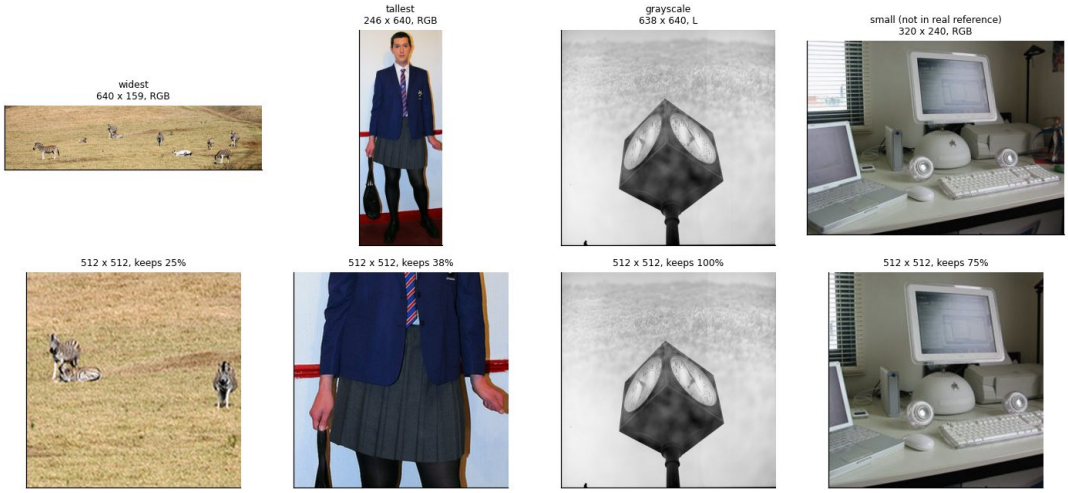


Figure 2. Before and after preprocessing for the widest, tallest, grayscale and smallest photos. Nothing is stretched; very wide photos lose their edges, which Section 8 checks against the object labels.

6. Data checks (counted, never changed)

Check	Result	Meaning
Photos with an EXIF rotation tag	0	No rotation decision is needed for this split
Near-duplicate photo pairs (hash distance ≤ 12 of 256 bits)	0	No duplicates inflate the real reference set

Check	Result	Meaning
Captions with a repeated word ("on the the street")	33	Annotator typos, kept as written
Captions without final punctuation	6,234	Common; kept, since it changes the prompt
Captions starting in lower case	3,708	Kept as written
Captions under 5 words, or with non-ASCII characters	0, 0	No unusable captions

7. Frozen splits

Split	Contents	Used for
eval_2000	2,000 eligible photos, one random caption each, seed 2026	All reported numbers in later milestones
eval_500	First 500 rows of eval_2000	Variance across seeds only, never tuning
dev_500	500 eligible photos outside eval_2000, seed 2027	Choosing guidance, scheduler, steps
real_ref_b_2000	2,000 eligible photos outside eval_2000, seed 2028	Real-against-real FID baseline at the same N

Overlap checks passed: neither dev_500 nor real_ref_b_2000 shares a photo with the test split, and redrawing the test split reproduces it exactly. dev_500 and real_ref_b_2000 share 353 photos, which is expected and harmless: the reference set contributes only pixels, never captions.

8. Object labels for every test caption

What was done. Each test caption was mapped to the COCO object types it names, then each named object was checked against the photo's boxes after the crop. A mention becomes a **target** (an object later images are checked for) if the word is unambiguous, the object is labelled in the photo, at least half its box survives the crop, and the kept part covers at least 0.5% of the image. The mapping rules passed all built-in checks ("teddy bear" is not a bear, "hot dogs" are not dogs, "two men riding bikes" names person and bicycle, "an orange cat" names no fruit).

Measure (eval_2000)	Count
Captions naming at least one COCO object	1,793 of 2,000
Captions with at least one target	1,692 (1,314 with a target other than person)
Object mentions found	2,768
Targets	2,291
Mentions excluded as unclear words	249
Named but not labelled in the photo	192
Excluded by the crop check	142
Most common targets besides person	dog 56, bus 55, cat 53, train 49, pizza 47, car 45, horse 42, elephant 42

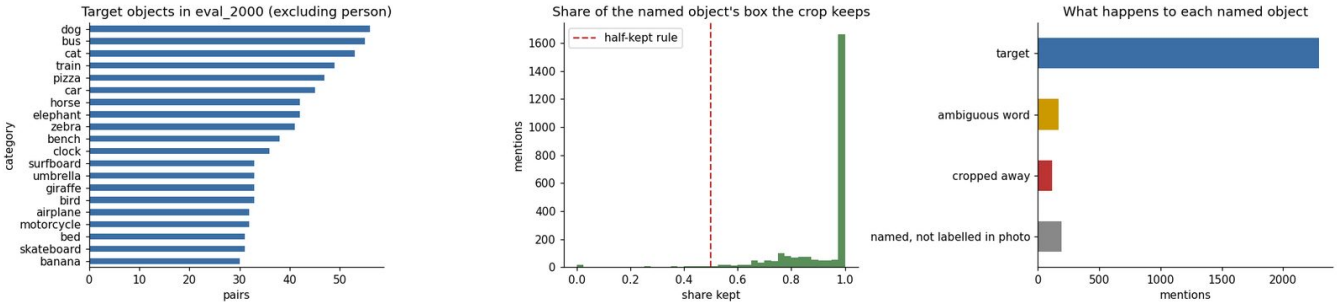


Figure 3. Left: most common targets. Middle: share of each named object's box that survives the crop; most objects keep all of it. Right: what happens to every named object.



Figure 4. Named objects excluded by the crop check (yellow: the kept square; red: the object).

What the outputs show. The crop check excludes two kinds of objects: ones the crop really cuts off (the first phone, 0% kept; the vase, 28%; the woman, 43%) and ones that survive but are tiny after the crop (the second phone, the cars and the distant people, each under 0.5% of the image). The middle histogram shows few objects lose more than half their box, so most of the 142 exclusions are tiny objects that no detector could fairly be asked to find. The most frequent unmapped words are not objects (sitting, two, white, standing, street), so the mapping is not missing common object words. "Tennis" (as in "tennis player") and "computer" (which could be a laptop, monitor or keyboard) are left unmapped on purpose.

9. Stress prompts

200 templated prompts in five families of 40, built from COCO objects so they can be judged automatically, each stored with its expected answer. Word-order pairs use exactly the same words in a different order (checked).

Family	Example	Expected answer stored
Word order (20 pairs)	a bear following a sheep on a grassy hill / a sheep following a bear …	subject, object, verb
Counting	three zebras	count per object
Colour binding	a yellow book and a white truck	colour per object
Spatial relations	a bowl to the left of a zebra	relation, first, second
Three objects	a horse, a bus and a bottle	objects that must appear

10. Words the CLIP tokenizer splits

CLIP splits rarer words into several tokens. Among the 2,000 test captions, none exceeds the 77-token limit (median 13 tokens, longest 36), but **379 captions (19%) contain a content word split into pieces**, most often computer (34), surfboard (26), zebras (22), hydrant (18), giraffes (12) and racquet (10). 24 of the split words name targets, including surfboard, zebras, giraffes, hydrant, skateboarder and suitcases. Any word-by-word analysis of attention must join these pieces into one word, or results for exactly these objects would be wrong.

11. Text → embedding

What was done. Every test caption was encoded with Stable Diffusion 1.5's own tokenizer and text encoder (CLIP ViT-L/14 text transformer, 123M parameters, 12 layers, width 768), one caption at a time with no attention mask, exactly as the pipeline encodes a prompt. The result for each caption is a 77 × 768 matrix: one 768-number vector for the start token, each caption token, the end token and the padding up to 77 positions. The empty prompt was encoded too, since classifier-free guidance runs every step against it.

Check	Result	Meaning
Cache shape and size	2,000 × 77 × 768 float16, 237 MB, 18.0 s	Fast enough to rebuild; every later run reads the same states
Cached row vs fresh encoding	identical	Caching changed nothing

Check	Result	Meaning
Scoring CLIP's text tower vs Stable Diffusion's text encoder	max difference 0.0000	CLIPScore will judge text with the same reader that generated the image
Caption → photo retrieval among 2,000 photos	R@1 46.8%, R@5 73.3%, R@10 84.0%, median rank 2	The right photo is usually first or second; chance R@1 is 0.05%

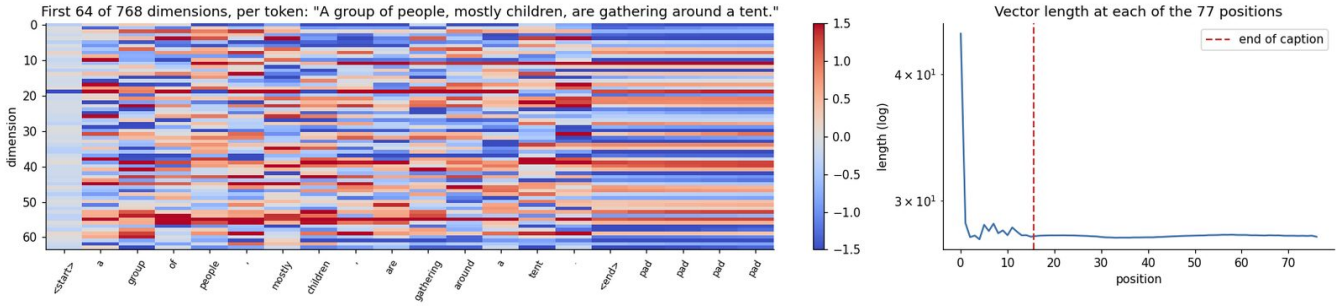


Figure 5. One caption's matrix (first 64 of 768 dimensions) and the length of the vector at every position.

What the outputs show. Each word gets a distinct pattern. The start token's vector is about one and a half times longer than a word's and almost flat in these dimensions: it is the attention sink that absorbs most of the model's attention. The padding positions are not empty: their columns repeat the end token's pattern, a summary of the whole caption, and because the pipeline uses no attention mask the model reads them too. This is why padding cannot be treated as blank when analysing what the model attends to.

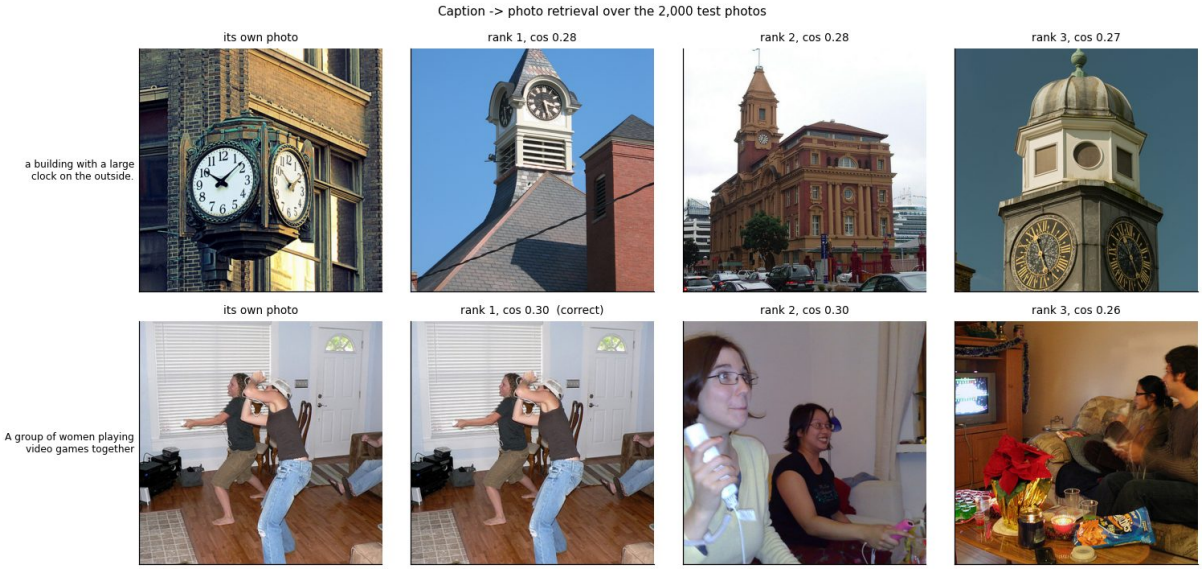


Figure 6. Retrieval examples: for two captions, the photo they came from and the three photos CLIP ranks highest.

A retrieval score depends on how many photos are searched: the same embeddings score higher among fewer candidates. These numbers are for 2,000 photos and should be compared only with numbers at the same pool size.

12. Stable Diffusion: first samples

Parity check. One caption and one seed were painted twice, from the raw text and from the cached 77 × 768 states. **The mean pixel difference is 0.000**: the two images are identical, so the embedding pipeline built here is exactly what drives the model. One 30-step image takes 4.7 s on a T4.

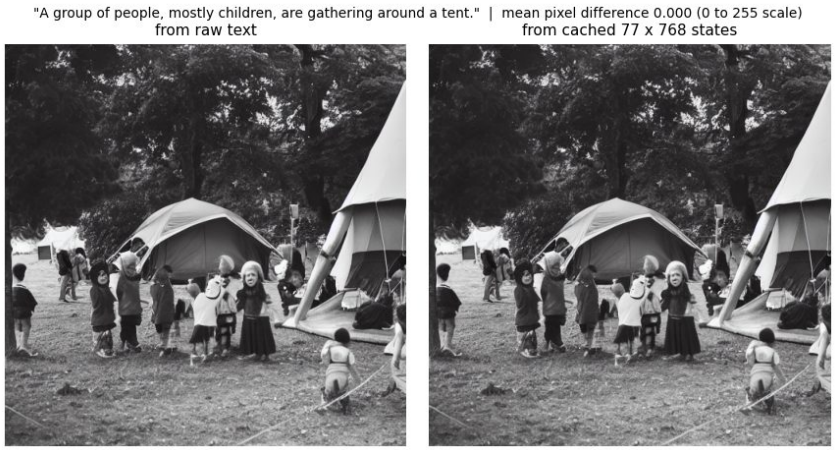


Figure 7. The same caption and seed painted from raw text (left) and from the cached embeddings (right).

Sample runs. Five test captions, chosen with a fixed seed among photos showing at least two kinds of object, three seeds each, generated from the cached embeddings with Stable Diffusion's defaults (PNDM, 30 steps, guidance 7.5): 15 images at 5.1 s each, saved with their caption, seed and settings in samples.csv.

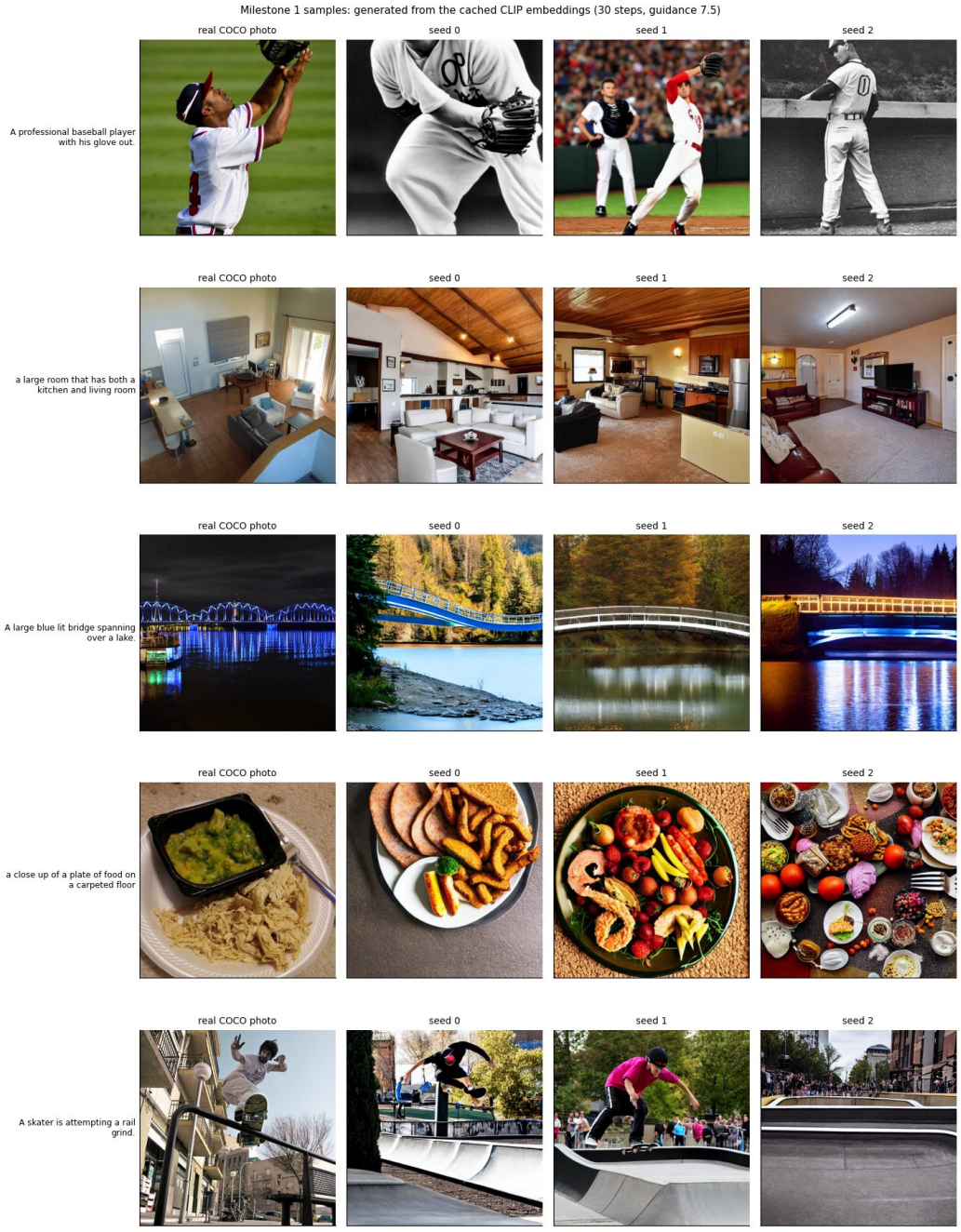


Figure 8. Milestone 1 samples: the real COCO photo, then seeds 0 to 2.

Caption	What the samples show
A professional baseball player with his glove out.	Player and glove appear in all three. Two seeds drift to black-and-white, a style the caption never asked for.
a large room that has both a kitchen and living room	All three show an open-plan kitchen and living room: the strongest match.
A large blue lit bridge spanning over a lake.	Bridge and lake appear in all three, but only seed 2 is lit at night; seeds 0 and 1 read "blue lit" as blue paint in daylight.
a close up of a plate of food on a carpeted floor	Food appears in all three, but the "carpeted floor" context is weak or missing, and seeds 1 and 2 show a spread of many dishes, not one plate.
A skater is attempting a rail grind.	Seeds 0 and 1 show a skater jumping at a skate park, not grinding a rail; seed 2 shows the skate park with no skater at all.

What the samples show. Objects and scenes named in the caption usually appear. Actions (a rail grind), lighting and attribute details ("blue lit"), context ("carpeted floor") and sometimes the main subject itself (seed 2 of the skater) are lost. This is exactly what the object labels and stress prompts were built to measure in later milestones.

Part 2 · Text encoders compared without training

13. Question and setup

Stable Diffusion 1.5 reads prompts through CLIP's text transformer. The project also plans to try BERT and GPT-2, which need a trained connector before Stable Diffusion can read them. Before spending GPU hours on that, Part 2 measures what each encoder's embeddings **already** capture, on the same 2,000 test captions. Nothing is trained.

Representation	Model	Trained on	Sentence vector
CLIP ViT-L/14	Stable Diffusion 1.5's own text encoder (reused from Section 10)	Image–text pairs	Mean of the caption's token states
CLIP (EOS)	Same model	Image–text pairs	The end-of-text token, the vector CLIP was trained to use
BERT-base	google-bert/bert-base-uncased	Text only (masked words)	Mean of token states
T5-base	google-t5/t5-base, encoder only	Text only (span filling)	Mean of token states
GPT-2	openai-community/gpt2	Text only (next word)	Mean of token states
TF-IDF words	Word counts weighted by rarity	—	Knows only which words appear

All four encoders are 768 numbers wide, so differences come from what they learned, not their size. Special and padding tokens are left out of every mean, the same rule for all. Token counts are similar (median 13 for CLIP and BERT, 14 for T5, 11 for GPT-2; no caption exceeds 41 tokens), so no encoder is disadvantaged by truncation. Embedding all captions with all four encoders took 101 s.

14. Test data

Paraphrases: for each test caption, another human caption of the same photo. **Scene labels:** the COCO supercategory of the photo's largest labelled object, taken from the boxes, not the caption (person 598, vehicle 326, furniture 268, animal 267, outdoor 108, indoor 94, food 89, others smaller). **Triplets:** three sentences with exactly the same words, built from the Milestone 1 stress prompts (20 word-order, 40 colour), e.g. "a bear following a sheep on a grassy hill" / "on a grassy hill a bear following a sheep" (same meaning, reworded) / "a sheep following a bear on a grassy hill" (meaning changed).

15. How similar are the encoders? (linear CKA)

Linear CKA compares two sets of embeddings of the same captions: 1 means the same information arranged differently, 0 means nothing shared.

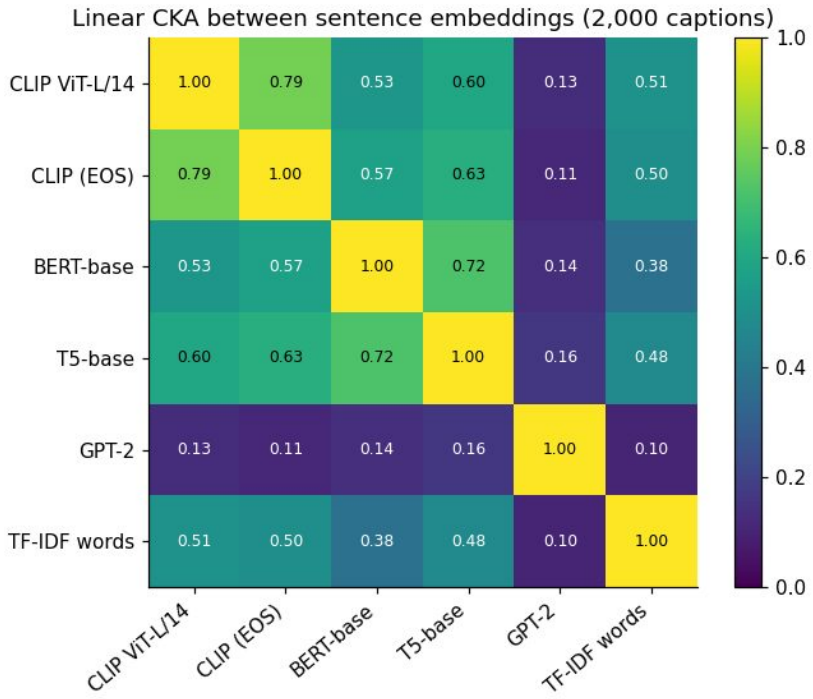


Figure 9. Linear CKA between every pair of representations on the 2,000 test captions.

Observations. (1) BERT and T5 agree with each other (0.72) more than either agrees with CLIP (0.53 and 0.60): the text-only models share a view of language that the image-trained model does not. (2) T5 is closer to CLIP than BERT is, so a T5 connector has less to change than a BERT connector. (3) BERT relates to CLIP about as much as plain word counts do (0.53 vs TF-IDF 0.51). (4) GPT-2 shares almost nothing linear with any other representation (0.10–0.16), explained in Section 17.

16. Do paraphrases land close together?

For each test caption, the 2,000 "other captions" are ranked by cosine similarity; the right answer is the other caption of the same photo. People describe one photo with different words, so this rewards meaning over wording.

Representation	R@1	R@5	R@10	Median rank
CLIP (EOS)	34.0%	57.9%	69.3%	3
CLIP ViT-L/14 (token mean)	21.4%	41.2%	52.2%	9
TF-IDF words	20.0%	38.0%	47.9%	12
T5-base	15.5%	31.0%	40.0%	22
BERT-base	13.0%	27.7%	35.1%	29
GPT-2	3.4%	9.5%	13.5%	315

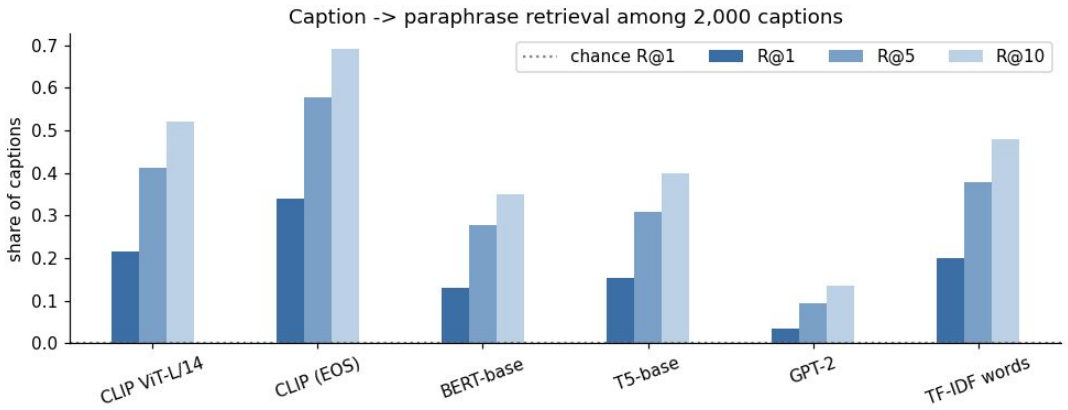


Figure 10. Caption → paraphrase retrieval among 2,000 captions (chance R@1 is 0.05%).

Observations. Only CLIP beats plain word counting. CLIP's own sentence vector (EOS) is far better than the mean of its token states (34.0% vs 21.4%), as expected, since EOS is the vector CLIP was trained on. BERT and T5 **lose to TF-IDF**: their mean-pooled vectors were never trained to compare sentences, and are known to be weak at it without sentence-level fine-tuning. This does not mean they carry no information; it means it is not arranged for similarity, which a trained connector could rearrange.

17. Do captions about the same kind of scene cluster?

Each caption is classified by a vote among its 10 nearest captions, leaving itself out; the label is the photo's scene type. The majority class (person) alone would score 29.9%.

Representation	10-NN scene accuracy
CLIP (EOS)	74.5%
CLIP ViT-L/14 (token mean)	70.0%
T5-base	68.5%
BERT-base	65.8%
TF-IDF words	64.4%
GPT-2	56.0%
Majority class	29.9%



Figure 11. t-SNE maps of the 2,000 test captions for each representation, coloured by the photo's main object type.

Observations. Every representation groups captions by scene well above chance, and the ranking matches retrieval: CLIP first, then T5, BERT and TF-IDF close together, GPT-2 last. The maps show **different kinds of structure**: CLIP forms many small, tight islands (captions about one specific kind of scene, such as giraffes or pizza), BERT and T5 form broad smooth regions (animals, vehicles, food) with softer borders, and GPT-2 shows almost no structure. GPT-2's mean-pooled vectors are nearly identical for every caption (similarity close to 1.000 between any two, Section 18); this is a known property of GPT-2's hidden states, which are dominated by a few very large dimensions. It explains GPT-2's weak retrieval and low CKA, and means a GPT-2 connector would at least need to normalise its inputs.

18. Word order and colour binding

For each triplet, an encoder **passes** if the same-meaning rewording is closer to the original than the meaning-changing swap. TF-IDF cannot tell them apart and scores exactly 50%.

Representation	Word order: pass rate	Colour: pass rate	cos(orig, swapped): word order	cos(orig, swapped): colour
CLIP ViT-L/14 (token mean)	0%	0%	0.914	0.761
CLIP (EOS)	0%	42.5%	0.968	0.960
BERT-base	0%	0%	0.993	0.996
T5-base	0%	0%	0.979	0.980
GPT-2	0%	3.8%	1.000	1.000
TF-IDF words	50% (tie)	50% (tie)	1.000	1.000

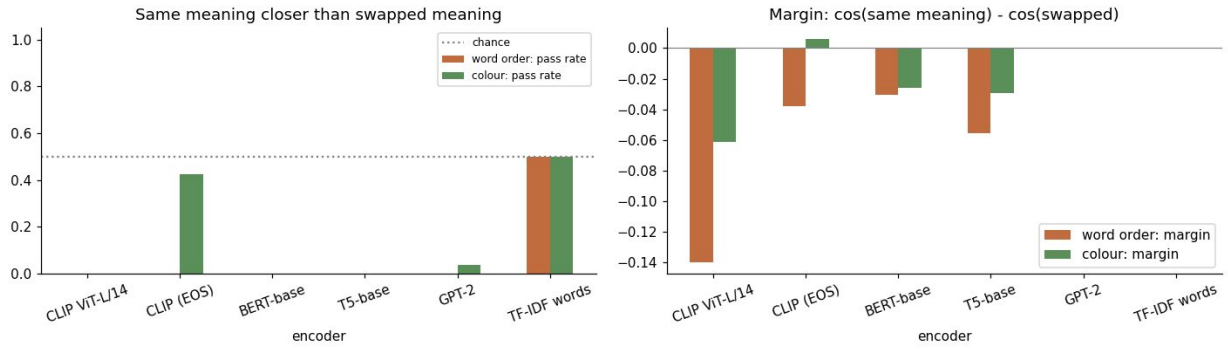


Figure 12. Pass rate and margin, cos(same meaning) – cos(swapped), for each representation.

What the outputs show, honestly. Every encoder scores 0% on word order and almost 0% on colour, below the 50% of word counting. This is mostly a **flaw in how the test was built**, not proof that every encoder fails: the "same meaning" rewording moves a whole phrase to the front of the sentence (or swaps the two objects), which changes the sentence's structure much more than the meaning-changing version, where only two words trade places. Sentence vectors from all these models follow surface structure strongly, so the structurally closer sentence wins every time. The test therefore measures structure, not understanding, and should not be used to choose a connector.

What can still be read from it. The similarity between a sentence and its role- or colour-swapped twin shows how much each representation changes when meaning changes but the words do not. BERT (0.993–0.996), T5 (0.98) and GPT-2 (1.000) sentence vectors barely move, so averaged over the sentence they hardly register who does what or which colour goes where. CLIP's token states move the most (0.91 for word order, 0.76 for colour), because CLIP reads left to right and every token after a swap sees a different context. CLIP's own sentence vector (EOS) gets colour binding right only 42.5% of the time, close to a coin flip, which agrees with published findings that CLIP behaves much like a bag of words.

Fix for the next run. Rebuild the triplets so the same-meaning version keeps the sentence structure and changes one word for a synonym ("a bear following a sheep" → "a bear trailing a sheep"), so structure is equal on both sides and only meaning differs.

19. What this means for the connector experiment

Decision	Evidence
Always train the **CLIP connector** as the control	Needed for a fair comparison; CLIP is also the best representation on every meaning test
Train the **BERT connector**	Required by the assignment and the proposal. Expect it to start behind CLIP: CKA 0.53, and it loses to word counting on paraphrases
Prefer **T5** as the third connector	Closer to CLIP than BERT (CKA 0.60), better than BERT on paraphrases (15.5% vs 13.0%) and on scene clustering (68.5% vs 65.8%)
GPT-2 is lowest priority	Nearly identical vectors for every caption, CKA 0.10–0.16, 3.4% paraphrase R@1; a connector would need to normalise its inputs first
Do not choose by the word-order test yet	The triplet design is confounded by sentence structure (Section 18); rerun with synonym rewordings first

A connector reads every token state, not one sentence vector, and learns to rearrange them, so these results describe the starting point a connector has to work from, not the images it will produce. The connector experiment, scored on the frozen splits, is what tests the images.

20. Limitations

• Sentence vectors are means of token states for BERT, T5 and GPT-2; models fine-tuned for sentence similarity would score higher. • The word-order and colour triplets are confounded by sentence structure, as explained in

Section 18. • Scene labels come from the photo's largest labelled object, which a caption may not mention. • Part 1's object labels come from a hand-built word table and have not been hand-checked; its crop-check exclusions mix objects cut off by the crop with objects too small to judge. • Five sample captions are a qualitative check, not a measurement.

21. Files produced and reproducibility

File	Contents
milestone1_summary.json	Every Part 1 number, plus Part 2's results under text_encoders
config.json, manifest.jsonl, captions.csv, preprocess_log.csv	Settings, and one row per image, caption and preprocessed photo
splits/, groundtruth_*.csv, stress_prompts.csv	Frozen splits, object labels, stress prompts
samples/ and samples.csv	15 sample images with caption, seed and settings
text_encoders/: A_cka.csv, A_paraphrase_retrieval.csv, A_knn_scene.csv, A_word_order_colour.csv, text_encoders_summary.json	Part 2 results
figures/	Every figure in this report

Environment: Google Colab, Tesla T4, PyTorch 2.11, Diffusers 0.42.1, Transformers 5.22. Every random draw uses a fixed seed. Rerunning Part 1 reproduced every figure and sample image of an earlier run pixel for pixel.

22. Conclusion

Part 1 completes every Milestone 1 notebook task: the dataset is chosen, cleaned and preprocessed; the environment is set up; the text → embedding pipeline reproduces Stable Diffusion's own images exactly; and the sample runs are done, with splits, object labels and stress prompts ready for fair evaluation later.

Part 2 shows that the text encoders are not interchangeable. CLIP, trained with images, captures the meaning of captions best; the text-only encoders, used as they are, organise captions differently (T5 closer to CLIP than BERT), do no better than word counting at matching paraphrases, and their sentence vectors barely change when the roles or colours in a sentence are swapped. A BERT or T5 connector therefore starts from a real gap that training has to close, which makes the planned CLIP-vs-BERT comparison meaningful, and points to T5 rather than GPT-2 as the stronger third candidate.